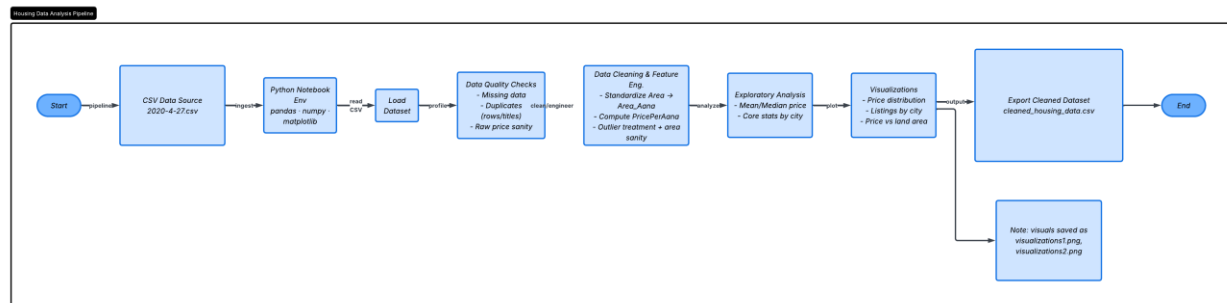


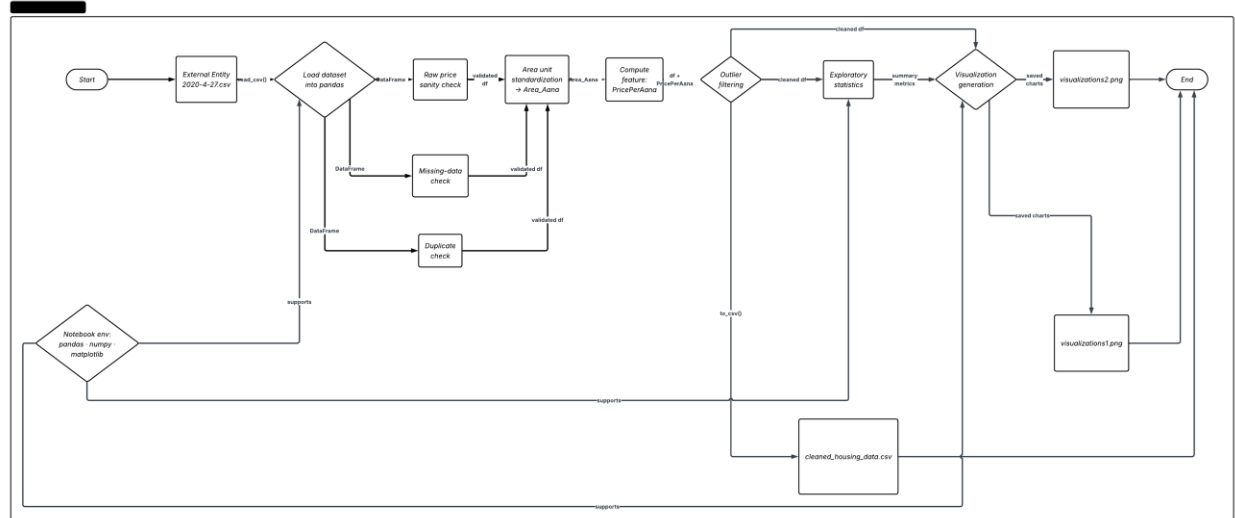
UML Diagrams: -

System Architecture Diagram: -



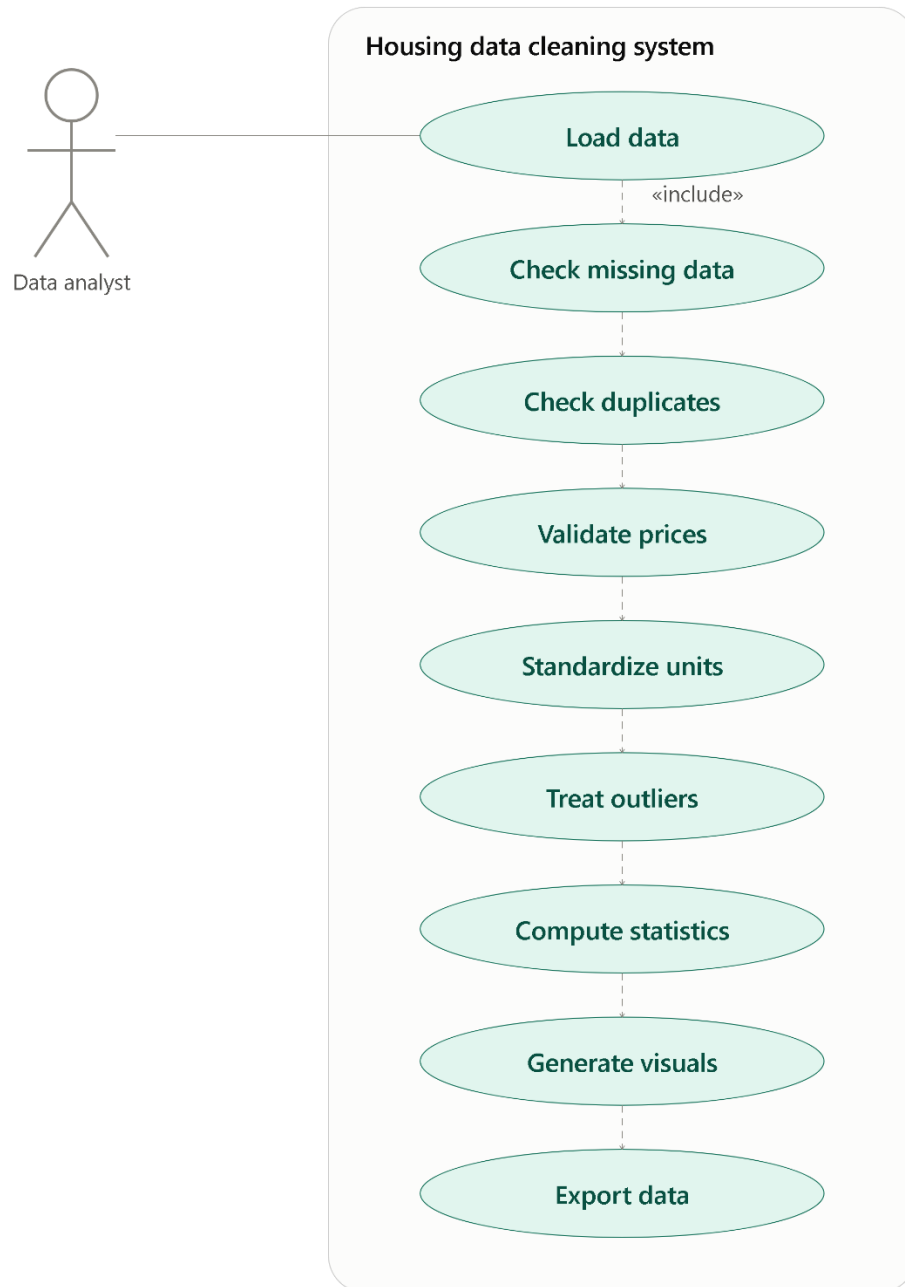
The Housing Data Analysis Pipeline begins by loading the raw housing dataset (2020-4-27.csv) into a Python notebook using the Pandas, NumPy, and Matplotlib libraries. After the dataset is loaded, the system performs data quality checks to identify missing values, duplicate records, and unrealistic price values that may affect the analysis. The data then undergoes cleaning and feature engineering, where different land measurement units (such as Aana, Ropani, Dhur, and Square Feet) are standardized into a single unit (Aana), a new feature called Price per Aana is calculated, and outliers or invalid records are removed to improve data quality. Once the dataset has been cleaned, exploratory data analysis is carried out to calculate important statistics, including the mean and median house prices and city-wise price comparisons. The system then generates visualizations, such as a price distribution histogram, a bar chart showing the number of listings by city, and a scatter plot illustrating the relationship between land area and house price. Finally, the cleaned dataset is exported as cleaned_housing_data.csv, while the generated charts are saved as visualizations1.png and visualizations2.png, completing the data analysis process.

Data Flow Diagram: -



The dataset is loaded into a Pandas DataFrame. After loading, the system performs several data validation checks, including a raw price sanity check to detect unrealistic property prices, a missing-data check to identify incomplete records, and a duplicate check to find repeated property listings. Once the dataset is validated, the land area values are standardized into a common unit (Area_Aana) by converting measurements such as Aana, Ropani, Dhur, and Square Feet into a single format. The system then computes a new feature called PricePerAana, which represents the price of a property per unit of land area. Next, outlier filtering removes records with unrealistic prices or land areas, producing a cleaned dataset suitable for analysis. The cleaned data is then used for exploratory statistical analysis, where summary measures such as mean price, median price, and city-wise statistics are calculated. Based on these results, the system generates visualizations, including the distribution of house prices, the number of listings by city, and the relationship between land area and property price. Finally, the cleaned dataset is exported as cleaned_housing_data.csv, while the generated charts are saved as visualizations1.png and visualizations2.png. The process ends after successfully producing these cleaned data and visualization outputs, which are ready for further analysis or reporting.

Use case: -



This Use Case Diagram illustrates the main functions of the Housing Data Cleaning System and the interaction of the Data Analyst with the system. The process begins when the data analyst loads the housing dataset into the system. The system then performs a sequence of data preprocessing tasks, including checking for missing values, detecting duplicate records, validating house prices, standardizing different land area units into a common format, and treating outliers to improve data quality. After cleaning, the system computes statistical summaries, generates visualizations such as charts and graphs, and finally exports the cleaned dataset for further analysis or reporting. The <<include>> relationship indicates that loading the data is a required step before any other data cleaning and analysis operations can be performed.